

Multiple Choice (10 marks)

1. The boolean expression $a[i] == \max \parallel \neg(\max \neq a[i])$ can be simplified to
 - (a) $a[i] == \max$
 - (b) $a[i] \neq \max$
 - (c) $a[i] < \max \parallel a[i] > \max$
 - (d) FALSE
2. In Python 3, let $r = \text{lambda } q: q * 2$. What is $r(3)$?
 - (a) 2
 - (b) 6
 - (c) 3
 - (d) 1
3. In python 3, which of the following is floor division?
 - (a) /
 - (b) //
 - (c) %
 - (d) |
4. Which of the following is NOT a reasonable justification for choosing to busy-wait on an asynchronous event?
 - (a) The wait is expected to be short.
 - (b) A busy-wait loop is easier to code than an interrupt handler.
 - (c) There is no other work for the processor to do.
 - (d) The program executes on a time-sharing system.
5. If a , b , and c are integers, which of the following conditions is sufficient to guarantee that the expression $a < c \parallel a < b \ \&\& \neg(a == c)$ evaluates to true?
 - (a) $a < c$
 - (b) $a < b$
 - (c) $a > b$
 - (d) $a == b$

True / False (10 marks)

Answer True or False

6. True or False: Python variable names are not case-sensitive, making 'Variable' and 'variable' interchangeable in the code.
7. True or False: A function that counts the values in a column or row is the best method for detecting data entry errors.
8. True or False: The regular expression $a^*(c + d)^* + b(c + d)^*$ describes the same set of strings as $(a^* + b)^*(c + d)$.
9. True or False: In Python 3, using a negative index such as 'abc'[-1] results in an error due to invalid indexing.
10. True or False: In Python 3, the sum of the list $l = [1, 2, 3, 4]$ is equal to 10.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to remove the duplicates from the given tuple.
12. Write a python function to left rotate the string.

End of Paper